



Charlie (Zhaohua) Qi

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PROFESSIONAL SUMMARY

Final-year Master of Computer Science student specialising in Artificial Intelligence, with portfolio projects across RAG systems, machine learning, NLP, computer vision, cryptography, databases and Python software engineering. Built public projects including a FastAPI/Streamlit regulatory knowledge assistant using PostgreSQL/pgvector, OpenAI embeddings and LangGraph; Seq2Seq text generation; CNN classification and segmentation; graph search; and a Python PKI certificate system. Brings 10 years of technical experience across biotech/nutrition product R&D, infant formula manufacturing and healthcare nutrition, with transferable strengths in automation, data handling, business systems, process improvement, technical documentation, stakeholder support and practical problem solving. Demonstrates learning agility and commitment to career transition through first-in-class results in three computer science units and a High Distinction average.

TECHNICAL SKILLS

AI / RAG

OpenAI embeddings, vector search, LangGraph, grounded answers, citations

Python Apps

FastAPI, Streamlit, REST APIs, pytest, Docker Compose

Data

SQL, PostgreSQL, pgvector, pandas, data preprocessing, query history

Automation

VBA tools, HTML calculator, SAP/M3 exposure, process improvement

ML / NLP / CV

PyTorch, TensorFlow/Keras, scikit-learn, Seq2Seq, CNNs

Security

RSA, AES, X.509, CSRs, CRLs, PKI concepts

AI / SOFTWARE PROJECTS

AI Regulatory Knowledge Assistant

[GitHub](#)

RAG, FastAPI, PostgreSQL/pgvector

- Built a document-grounded RAG assistant using FastAPI, Streamlit, PostgreSQL/pgvector, OpenAI embeddings and chat generation to answer questions over local regulatory documents with citations and query history.
- Implemented TXT/PDF/DOCX ingestion, chunking, vector retrieval, LangGraph orchestration, Docker Compose setup and simple CSV plus optional RAGAS evaluation workflows.

Python PKI Certificate System

[GitHub](#)

Python, cryptography, X.509

- Implemented an educational PKI simulator with Root CA, Sub-CAs, clients, encrypted certificate requests, validation and revocation.

Seq2Seq Recipe Generation with Attention

[GitHub](#)

NLP, PyTorch

- Built a PyTorch LSTM encoder-decoder recipe-generation system using ingredient lists as source sequences and recipe instructions as targets.

Computer Vision Classification and Segmentation

[GitHub](#)

CV, TensorFlow/Keras

- Implemented Harris corner detection and Canny edge detection, comparing custom outputs with library baselines.

AI Graph Search Algorithms

[GitHub](#)

Python, algorithms, CLI

- Implemented BFS, DFS, Greedy Best-First Search and A* over weighted CSV graph inputs with a reusable package API, command-line interface and focused tests.

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Biotech product R&D, manufacturing and healthcare experience

BIOTECH PRODUCT R&D, MANUFACTURING AND SYSTEMS

Technical Manager - Infant Milk Formula

Melbourne | Apr 2022 - Present

Bellamy's Organic

- Lead infant formula and nutritional product development from concept to launch, coordinating marketing, sales, suppliers, regulatory stakeholders and third-party manufacturers.
- Manage formulation assessment, nutrition panels, ingredient sourcing, packaging validation, sensory evaluation, shelf-life validation and controlled technical documentation.
- Support regulatory and market-access work including SAMR GB-2 registration, China organic WIT certification and FDA exemption-related technical evidence.
- Use SAP and cross-functional data inputs to support technical decisions, supplier coordination, product changes and operational follow-up.
- Developed an HTML ingredient forecasting calculator to support sales and procurement planning, improving business visibility and stakeholder alignment.

Formulation Technologist - Operation

Jan 2020 - Apr 2022

Fonterra Australia

- Supported new product/process development, drying and processing challenge resolution, equipment commissioning and campaign-management improvement in a manufacturing environment.
- Used SAP and production data to support formulation, scheduling and operational follow-up across manufacturing campaigns.
- Developed a production calculator to automate ingredient conversion calculations, reducing manual work and saving operational time.

Product Development Technologist - Nutritionals

Apr 2018 - Jan 2020

Bega Cheese

- Developed formulations, bills of materials, specifications, operating conditions and regulatory documentation across nutritional powder categories.
- Used planning and business systems including M3 to support product development, formulation changes and manufacturing readiness.
- Built a goat milk formulation calculator in VBA to automate manufacturing operation calculations and improve process consistency.

Dry Blending Formulator and QA Officer

Sep 2017 - Apr 2018

Australian Natural Milk Association

- Supported dry-blending formulation, GMP audits, supplier documentation, quality systems, corrective actions and product-compliance reporting.

HEALTHCARE EXPERIENCE

Clinical Dietitian Placement

2016

Sydney Hospitals

- Completed clinical nutrition placements at Concord Hospital, Sydney Prince Alfred Hospital, Bondi Mental Health Clinic and Sydney Westmead Children's Hospital.
- Used electronic patient record and hospital information systems to review clinical notes, nutrition history and care documentation.
- Supported paediatric/infant nutrition, allergies and intolerances, rehabilitation nutrition, food-safety auditing and clinical research activities.

EDUCATION

- **Master of Computer Science, Artificial Intelligence specialisation** | Monash University | Expected Jun 2026

- Academic achievements: First in class in Machine Learning, Image and Video Processing (Computer Vision), and Applied Practice 1; High Distinction average.

- **Master of Science, Nutrition and Dietetics** | The University of Sydney | 2016

- **Bachelor of Science, Food Science and Nutrition** | The University of Auckland | 2014

- Awards: First place in Food Quality Attribute; Certificate of Distinction in Statistics for top 10%.

COURSEWORK EVIDENCE

- Relevant coursework: NLP, Machine Learning, AI, Computer Vision, Cryptography, Databases, Software Engineering, Python and Applied Practice.
- Databases and Data Handling: conceptual and logical database design, relational schema design, SQL scripting, data preprocessing and exploratory analysis.
- Systems and Networking: routing, EIGRP metric analysis, static routes, firewall configuration, Wireshark packet analysis and TCP/IP concepts.